

Management of paediatric splenic injury in the New South Wales trauma system



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ABSTRACT

Since the 1980's, paediatric surgeons have increasingly managed blunt splenic injury (BSI) in children non-operatively. However, studies in North America have shown higher operation rates in non-paediatric centres and by adult surgeons. This association has not been examined elsewhere.

Objective: To investigate the management of BSI in New South Wales (NSW) children, to determine the patient and hospital factors related to the odds of operation. Secondly, to investigate whether the likelihood of operation varied by year.

Methods: Children age 0–16 admitted to a NSW hospital between July 2000 and December 2011 with a diagnosis of BSI were identified in the NSW Admitted Patient Data Collection, and linked to deaths data from Registry of Births Deaths and Marriages, and Bureau of Statistics. The operation rate was calculated and compared between different hospital types. Univariable analysis was used to determine patient and hospital factors associated with operative management. The difference in the odds of operation between the oldest data (July 2000–December 2005) and most recent (January 2006–December 2011) was also examined. Multivariable logistic regression with stepwise elimination was then performed to determine likelihood of operative management according to hospital category and era, adjusting for potential confounders.

Results: 955 cases were identified, with 101(10.6%) managed operatively. On multivariable analysis, factors associated with operation included age (OR 1.11, 95% CI 1.01–1.18, $p < 0.05$), massive splenic disruption (OR 3.10, 95% CI 1.61–6.19, $p < 0.001$), hollow viscus injury (OR 11.03, 95% CI 3.46–34.28, $p < 0.001$) and transfusion (OR 7.70, 95% CI 4.54–13.16, $p < 0.001$). Management outside a paediatric trauma centre remained significantly associated with operation, whether it be metropolitan adult trauma centre (OR 4.22 95% CI 1.70–10.52, $p < 0.01$), rural trauma centre (OR 3.72 95% CI 1.83–7.83, $p < 0.001$) or metropolitan local hospital (OR 5.23, 95% CI 1.22–18.93 $p < 0.05$). Comparing the 2 eras, the overall operation rate fell, although not significantly, from 12.9% to 8.7% (OR 1.3, 95% CI 0.89–243 $p = 0.13$).

Conclusion: While Paediatric Surgeons have wholeheartedly adopted non-operative management, away from paediatric centres, children in NSW are still being operated on for BSI unnecessarily. While the factors at play may be complex, further evaluation of the management and movement of injured children within the broad NSW trauma system is required.

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Introduction

Injury is one of the biggest health issues for children. Globally, it is the leading cause of death and morbidity over the age of one [1]. Injury remained the leading cause of mortality in New South Wales (NSW) children and adolescents in 2014, accounting for 37% deaths [2].

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Most injuries are due to blunt trauma and up to 13% have an abdominal injury [3]. The most commonly injured organ is the spleen accounting for up to 46.7% of all abdominal injuries [4]. Injury can range from a small sub-capsular haematoma to complete disruption of the organ, and it is often associated with multiple other injuries in and outside of the abdomen [5,6].

How children are managed within trauma systems significantly influences morbidity and mortality [7]. The way Blunt Splenic Injury (BSI) is managed exemplifies this [8]. According to Ein, and now widely accepted by paediatric trauma surgeons, regardless of the grade and associated injuries, “preservation of the spleen in blunt trauma should not be in formalin but instead in the peritoneal cavity” [9]. The driving reason behind this view is twofold. Firstly, avoiding splenectomy avoids Overwhelming Post Splenectomy Sepsis (OPSI) [10]. OPSI is a particular risk for children, occurring at a rate of 2–4% with a mortality of up to 50% [11]. Secondly, avoiding operation avoids all of the potential adverse sequelae of surgery. The goal of management in BSI is therefore spleen preservation along with avoidance of the complications of laparotomy.

Following the first reports of successful Non Operative Management (NOM) of paediatric BSI in the 1960's and 70's [12–14], along with the advent of cross-sectional imaging and introduction of paediatric trauma centres [1,15], NOM gradually became the norm. This culminated in the publication of evidence based guidelines by the American Paediatric Surgical Association (APSA) in 2000 [16]. The benchmark for NOM is 89–95% for all BSI and 97–100% for isolated BSI [16,17].

While there has been a clear and almost universal adoption of NOM in paediatric centres, the operation rate across the North American health system is reported to remain high, and mainly involves children managed outside of paediatric trauma centres [6,8,18–20]. The factors at play may be complicated but include patient characteristics [6,18,21–28], hospital characteristics [6,8,18–21,29,30] and surgeon practices [23,30–34].

NSW is the most populous state in Australia and has three paediatric trauma centres (PTC's), 17 adult trauma centres, ten of which are regional or rural, and over 200 metropolitan or rural hospitals where injured children may also present. In 2011, Fick reviewed 69 cases of BSI managed in a single NSW PTC over 10 years and found an overall NOM rate of 91% [26]. However, the extent to which paediatric trauma management occurs outside of NSW paediatric trauma centres, and specifically, the extent to which the guidelines for NOM of BSI are observed, has not been studied.

This study aimed to describe the incidence and management of BSI in children aged 0–16 across the state of NSW, over the ten years following the publication of the BSI NOM guidelines by Stylianos [16], with particular attention to the patient and hospital factors that may be associated with operative management, and whether the odds of operation varied by year. We hypothesised that the odds of operation would be higher outside of major paediatric centres, in more severely injured children, and in earlier years. The findings will have implications for the wider management of children within the NSW trauma system.

The study was approved by both the NSW Population and Health Services and Aboriginal Health and Medical Research Council Ethics Committees.

Methods

The study was a retrospective cohort study from 1st July 2000 to 31 December 2011. Children aged 0–16 years with a diagnosis of Splenic Injury (SI) were included. Cases were identified from records within the Admitted Patient Data Collection (APDC). This database is maintained by the NSW Ministry of Health. It contains

information on all NSW hospital admissions, and contains data on demographics, facilities, diagnoses, external causes, procedures and destination on discharge. Diagnoses, external causes and procedures were coded according to the International Classification of Diseases, 10th revision, Australian Modification (ICD-10-AM). These records were linked to NSW mortality data contained in Registry of Births, Deaths and Marriages (RBDM) and the Australian Bureau of Statistics (ABS) death registration. Data was accessed through, and linked by, the NSW Centre for Health Record Linkage (CHeReL) and provided to researchers in a de-identified format.

Data was cleaned to remove duplicates, patients where date of death preceded admission or separation date, ages outside of the study range and cases where the external cause code indicated a penetrating injury, including those due to firearms ($n = 4$).

Outcome and explanatory variables used in the analysis are described in Table 1.

Outcomes

The primary outcome was management of children with BSI categorised as operative management (OM) or Non Operative management (NOM). OM was defined using the ICD-10-AM codes for laparotomy, with or without splenectomy or another spleen-preserving procedure such as partial splenectomy or splenorhaphy.

Explanatory variables

Explanatory categorical and continuous variables were extracted or created using raw data in the three datasets.

The main explanatory variable was the trauma centre type. Facilities were categorised according to the trauma centre designation assigned by NSW Ministry of Health Institute of Trauma and Injury Management (ITIM) as (1) paediatric trauma centre (PTC) –which was further subcategorised in to primary presentation and transfer to a PTC-, (2) metropolitan adult trauma centre (MATC), (3) rural trauma centre (RTC), (4) metropolitan local health facility (MLH) and (5) rural local health facility (RLH).

Total length of stay (LOS) in days was calculated by combining all the records for one period of stay, which may have included transfer between hospitals. Intensive care (ICU) length of stay was separately recorded in hours. Death within 30 days was obtained from the APDC mode of discharge code and correlated with the deaths data. Other categorical variables were formulated by grouping relevant date, demographic, facility, diagnostic, external cause and procedure codes. The ABS Socio-economic Index for Advantage and Disadvantage (SEIFAD) classifies NSW localities into 10 deciles, with the first decile representing the most socioeconomically disadvantaged and the least advantaged and the 10th representing the least disadvantaged and the most advantaged. SEIFAD was assigned according to postcode. External causes, injuries and procedures, including operations and transfusion, were grouped using the ICD-10-AM codes. External causes were grouped in to common mechanism categories. While spleen injury grading according to the American Association for the Surgery of Trauma is not coded in the APDC, spleen injuries were grouped in to mild (haematoma or capsule tear) equating to grade I–III, moderate (laceration extending in to parenchyma) equating to grade II–IV, severe (massive splenic disruption) equating to grade IV–V, and unspecified [35].

Associated major injuries were grouped according to body region. Minor soft tissue injuries and sprains/strains were excluded. Because mapping of Injury severity scores to ICD-10-AM was not available to us, injury severity was represented by the

Table 1

Summary of outcome and explanatory variables used in the regression analysis.

	Variable	Value
Outcome variable	operation grouped ICD-10-AM procedure codes	binary yes/no “yes” sub-grouped Laparotomy Splenectomy Spleen preserving procedure
Explanatory variables	hospital trauma centre designation – categorised NSW hospital facility codes based on NSW Ministry of Health, Institute of Trauma and injury Management designation	paediatric trauma centre (PTC) Initial transferred metropolitan adult trauma centre (MATC) rural trauma centre (RTC) metropolitan local health facility (MLH) rural local health facility (RLH)
	hospitalisation characteristics length of stay (LOS) Intensive care (ICU) length of stay death within 30 days	days – continuous hours – continuous binary yes/no
	demographics age gender Indigenous status socioeconomic index for advantage and disadvantage (SEIFAD)	continuous binary male/female binary yes/no decile 1–10
	mechanism categorised ICD-10-AM external cause codes. (only main ones listed)	motor vehicle occupant pedestrian agricultural vehicle including ATV falls motorcyclist pedal cyclist animal rider other
	injury severity – compound injury risk ratio (C-IRR)	continuous
	spleen injury severity ICD-10-AM diagnostic codes	capsule tear/haematoma (grade 1–III) parenchymal laceration (grade II–IV) massive splenic disruption (grade IV–V) other spleen injury
	associated major injuries categorised ICD-10-AM diagnostic codes	isolated blunt splenic injury intracranial liver kidney hollow viscus pancreatic other intraabdominal thoracic upper limb lower limb spinal/back pelvic fracture
	transfusion	binary yes/no
	era	era 1 - July 2000–December 2005 era 2 - January 2006–December 2011

compound injury risk ratio (C-IRR). The IRR is the basis of the ICD Injury Severity Score (ICISS) and has been calculated as a ratio of all survivors with a particular ICD-10-AM injury code, to all patients with that injury code including deaths. The C-IRR is the product of all the IRRs for a particular patient's coded injuries [36].

While physiological data such as blood pressure, pulse and blood test results, are not contained within the datasets, C-IRR and transfusion are presumed to correlate with physiological parameters. The entire cohort was analysed as a whole then subgroup analysis was performed on two eras – the oldest data (July 2000–

December 2005 – 5.5 years) and newest data (January 2006–December 2011 – 6 years).

Statistical analysis

All analyses were conducted using R statistical software version 3.2.3 or SAS statistical software version 9.3 (SAS Institute Inc., Cary, NC, USA).

Descriptive statistics for each variable were generated. The operation rate was calculated as a proportion of total admissions with a splenic injury and compared between different facility categories. Patient factors and facility category as well as era and their association with operative management were examined with univariable logistic regression, using the Wald z-test, generating odds ratio, 95% confidence limits and p value. For continuous explanatory variables, the odds ratio relates to each increment in the numeric value of the variable.

Multivariable logistic regression with stepwise elimination was then performed to determine likelihood of operative management according to hospital category, adjusting for the effect of potential confounders on the variables of interest. Total length of stay, ICU length of stay and death within 30 days were excluded from the multivariable analysis as they may be considered consequences rather than predictors of operation. To allow for possible lack of homogeneity within hospital categories, a Generalised Linear Mixed Effects Model was run, adding the individual hospital identifier as a random effect.

Variables found to be significant on univariable analysis to a p value of 0.25, were included in the initial model. Variables were eliminated based on maintaining the most robust model according to three criteria (i) the Hosmer Lemeshow test, (ii) the Akaike Information Criteria, and (iii) the chi-square on null versus deviance residuals.

Results

Hospital

Between July 2000 and December 2011, 955 children aged 0–16 were admitted to hospital in NSW with BSI. 369 (38.7%) had some or all of their management at one of 3 PTCs in the state (1.3% of NSW hospitals). Of these, 197 (20.6%) presented primarily to a PTC and the remainder (n = 197, 20.6%) were transferred to a PTC after a mean of 0.53 days (range 0–4). Almost 40% (n = 358, 37.5%) were admitted to an ATC. Just under 10% (n = 94, 9.8%) of these were to one of five MATCs and 264 (27.6%) to one of ten RTCs (7.2% of NSW hospitals). The remaining 243 (25.4%) were managed entirely at a non-trauma centre (91.5% of all NSW hospitals) including 216 (22.6%) who were managed at RLHs.

Demographics and external cause

The total number of admissions increased each year of age up to age 15 (Fig. 1). 80% were male, predominant at every age. 48/955 (5%) were Aboriginal or Torres Strait Islander, who make up 2.9% of the NSW population (<http://www.abs.gov.au/>). The most common external causes are shown in Fig. 2. Motor vehicle occupant or pedestrian increased in total number but reduced as a proportion of total injuries with age while injury from pedal cycles and motorcycles increased both in total and as a percentage with age (Fig. 3).

Associated injuries

545 (57.1%) of patients had isolated BSI. The most common associated injuries for the remaining were thoracic (n = 161; 16.9%),

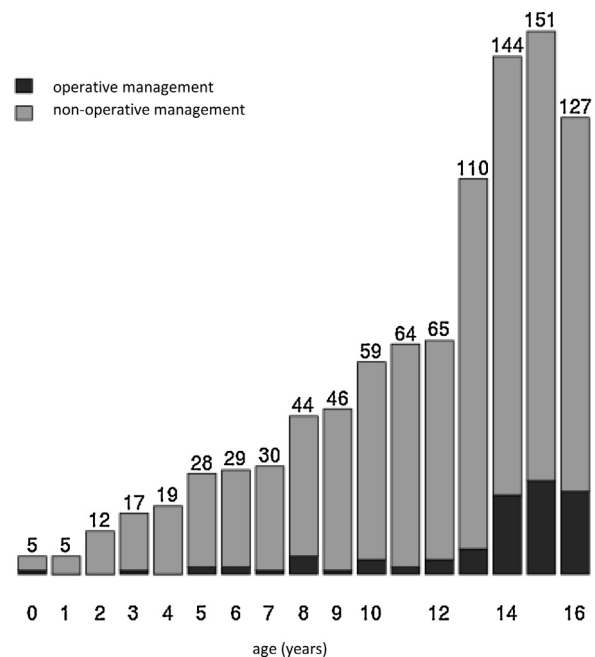


Fig. 1. The number of cases of blunt splenic injury and number of operations for each year of age.

and abdominal injuries including kidney (n = 99, 10.4%), liver (n = 66, 6.9%), pancreas (n = 20, 2.1%), hollow viscus (n = 23, 2.4%), and other abdominal injuries (n = 127, 13.3%). Orthopaedic injuries included upper limb (n = 99, 10.4%), lower limb (n = 42, 4.4%), back and spinal (n = 53, 5.5%), and pelvic fractures (n = 46, 4.8%). 37 (3.9%) had a co-existing significant head injury.

Most spleen injuries consisted of haematoma/capsule tear (n = 383, 40.1%). Parenchymal laceration accounted for 98 (10.3%), and massive splenic disruption, or grade V injury, for 257 (26.9%). The remainder (n = 217, 22.7%) were unspecified.

Management

In total, 101/955 (10.6%) were treated operatively. Of these, 59 (6.2%) underwent splenectomy, 26 (2.7%) underwent some sort of spleen preserving procedure (splenorrhaphy or partial splenectomy) and 16 (1.7%) laparotomy with no procedure on the spleen.

The mean LOS was 8 days, median 6. LOS was longer at PTC's (mean 11.2, median 6) than non PTC (mean 6.1, median 6) (p = 0.01).

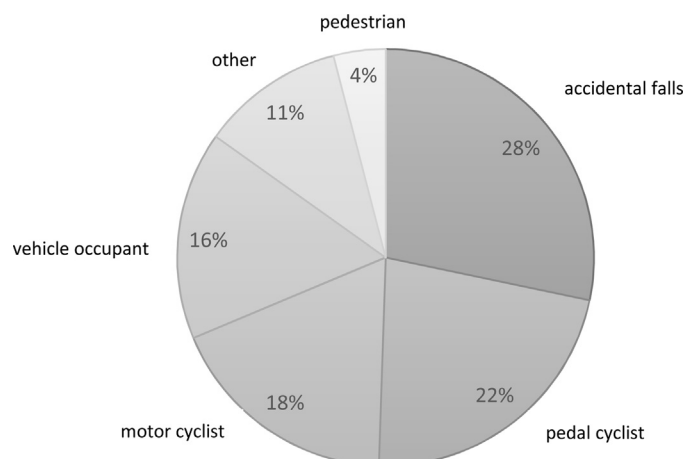


Fig. 2. Most common external causes of blunt splenic injury by percentage.

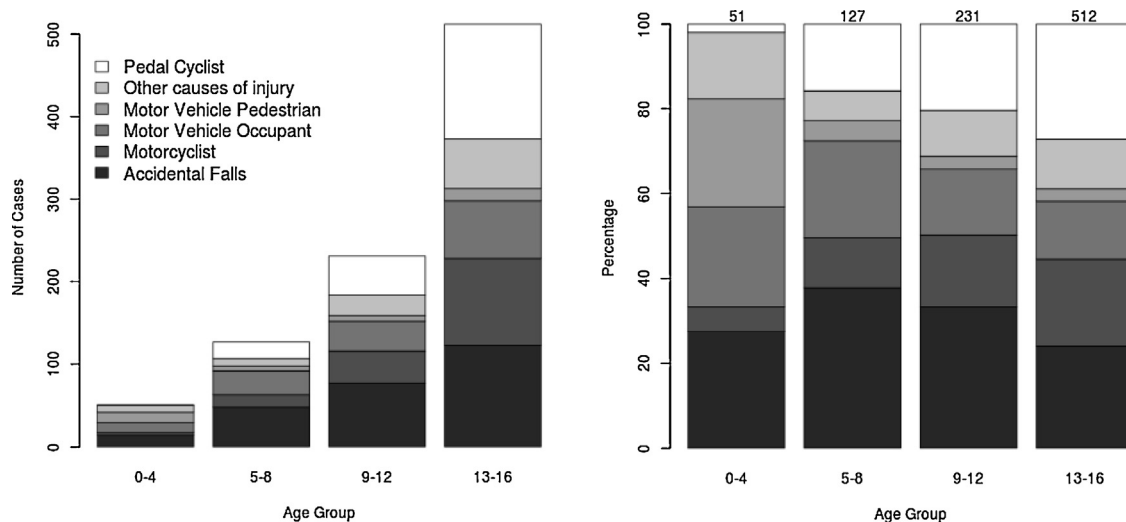


Fig. 3. Graphs showing change in total number (left) and relative percentage (right) of most common external cause of blunt splenic injury according to age.

Factors associated with operative management

Analysis of patient and hospital factors associated with operation is presented in Table 2. All causes 30-day mortality was 1.9%. The death rate was higher in the operative group 6/101 (5.9%) than the non-operative group (12/854, 1.4%, $p < 0.01$).

PTCs were the only facilities where benchmarks for NOM were met. On univariable analysis, operative management was significantly less likely in PTCs (5.6%) compared to non PTCs (13.5%), whether admission to a PTC was primary or the result of a transfer ($p = 0.001$). OM was also more likely in ATCs in both rural and metropolitan localities (14.4%, 18.1% respectively, $p < 0.001$) and in RLHs (10.2% $p < 0.05$).

As shown in Table 2, patient characteristics significantly associated with operation on univariable analysis included age, lower SEIFAD, overall length of stay and length of time in intensive care, severe grade of splenic injury, specific associated injuries, blood transfusion, and injury severity as indicated by the compound IRR. Operation was significantly less likely in isolated BSI and in those with mild grade. The likelihood of operation was not significantly related to gender (OR 0.96, 95% CI 0.58–1.66), Indigenous status (OR 0.55, 95% CI 0.13–1.54), moderate grade (OR 1.4, 95% CI 0.77–2.63), renal injury (OR 1.32, 95% CI 0.68–2.39), pancreatic injury (OR 2.16, 95% CI 0.61–6.02) or lower limb injury (OR 1.74, 95% CI 0.69–3.81).

Comparing the two eras, the overall operation rate fell from 12.9% to 8.7% ($P < 0.05$). Reduction occurred at both PTCs and non-PTCs however the reduction was not significant when facility categories were examined individually [PTC 8.0% in era 1 and 4.5% in era 2 ($p = 0.18$) and non-PTC 15.5% in era 1 down to 11.6% in era 2 ($p = 0.18$)]

The final multivariable regression model retained 13 variables (Table 2). The patient factors that remained significantly associated with operation included age (OR 1.09, 95% CI 1.01–1.18) massive splenic disruption (OR 3.31, 95% CI 1.61–6.19,) associated hollow viscus injury (OR 11.03, 95% CI 3.46–34.28) and blood transfusion (OR 7.70, 95% CI 4.54–13.16). Other associated injuries and overall injury severity did not influence the odds of operation

Most importantly, after adjusting for confounders, being managed away from a PTC remained significantly associated with operation, whether it be a MATC (OR 4.22, 95% CI 1.70–10.52) RTC (OR 3.672 95% CI 1.83–7.83) or MLH (OR 5.23 95% CI 1.22–18.93). There was no random effect observed from individual hospitals within hospital categories.

Discussion

This is the first study outside of North America to demonstrate disparities in management of paediatric BSI between various hospital types, using administrative data. Patient demography, mechanisms and associated injuries were similar to previously reported North American data [6,8,21,29], indicating that this NSW cohort is reasonably compared to those in North American studies. The overall NOM rate of 89.5% in NSW is just within the APSA guidelines [16]. However, despite the fact that haemodynamic instability is the sole indication to operation for splenic injury [37], the designation of the hospital a child is admitted to within the NSW Trauma System is independently associated with the chance NOM will be pursued. Importantly, the operation rate was significantly higher in non PTCs where overall the APSI benchmark is not met. After adjusting for confounders, the odds of operation were more than 4 times higher at metropolitan non-paediatric centres, regardless of whether it was a trauma centre or not, and 3.7 times higher at rural trauma centres. Only PTCs and rural non trauma hospitals met the NOM benchmark of over 89%. Importantly, the low operation rate in PTCs holds both for children admitted primarily, and for those transferred from another centre.

This disparity was first reported in North America in the late 1990's, when operation rates were as high as 45.6% in ATCs compared to 5.5% in PTCs [38–42]. Most North American population based and single centre reports over the last 15 years have shown a clear and almost universal direct practice of the Stylianos's 2000 APSA guidelines in PTCs, where operation rates have continued to fall and are now less than 3% with splenic salvage approaching 100% [6,8,19,21,29,30]. While operation rates have also fallen in North American adult centres, the risk of operation remains 2–10 times higher than at a PTC [8,18,19,43,44]. Our data indicates that this disparity is also evident in NSW, Australia. It is likely that in Australia, the handling of children within geographically diverse trauma systems, as well as patient and surgeon factors, play a part in explaining these findings as they do in North America [30].

Trauma systems and the influence on management of splenic injury

The designation of PTCs aims to improve trauma care in children. Indeed, PTCs have been confirmed to reduce morbidity and mortality in injured children, both in the US and in Australia [1,40,42,45,46]. However, in the US and Canada, geography and a

Table 2

Factors associated with operative and non-operative management in blunt splenic injury. Unadjusted variables with p-value of less than 0.25 are listed. These were then used in the multivariable regression model, for which results of retained factors are listed as adjusted results. SEIFAD = Socioeconomic index for advantage and disadvantage. BSI = blunt splenic injury. ICU = intensive care unit. ^ahospitalisation characteristics were included in the univariable analysis against operation but not in the multivariable analysis as they may be considered outcomes rather than explanatory of operation.

Effect	OM	NOM	Unadjusted results			Adjusted results		
			P value	OR	95% CI	P value	OR	95% CI
Total (%)	101 (10.6)	854 (89.4)						
Demographics								
age: mean (median)	13.34 (14)	11.61(13)	<0.001	1.17	1.09–1.26	<0.05	1.09	1.01–1.18
SEIFAD: mean (median)	4.35 (4)	5.06 (4)	0.01	0.90	0.82–0.97	0.10 (NS)	0.92	0.82–1.02
Hospital characteristics (reference paediatric trauma centre)								
paediatric trauma centre: 354 (37.1%)	20 (5.6)	334 (94.4)						
metropolitan adult trauma centre: 94 (9.8%)	17 (18.1)	77 (81.9)	<0.001	3.69	1.83–7.37	<0.01	4.22	1.70–10.52
rural trauma centre: 264 (27.6%)	38 (14.4)	226 (85.6)	<0.001	2.81	1.61–5.04	<0.001	3.72	1.83–7.83
metropolitan local health facility: 27 (2.8%)	4 (14.8)	23 (85.2)	0.07 (NS)	2.90	0.80–8.48	<0.05	5.23	1.22–18.93
rural local health facility: 216 (22.6%)	22 (10.2)	194 (89.8)	<0.05	1.89	1.01–3.59	0.09 (NS)	1.97	0.89–4.40
Hospitalisation characteristics^a								
length of stay days: mean (median)	15.20 (7)	7.23(5)	<0.001	1.023	1.01–1.04	not included		
ICU hours: mean	50.35	15.58	<0.05	1.002	1.001–1.004	not included		
death at 30 days N = 18 (1.9%)	6 (33.4)	12(66.6)	<0.01	4.43	1.51–11.69	not included		
Spleen injury characteristics (reference other spleen injury)								
capsule tear/haematoma: 383(40.1%)	15 (3.9)	368 (96.1)	<0.01	0.38	0.19–0.75	0.12 (NS)	0.54	0.24–1.17
parenchymal laceration: 98(10.3%)	14 (14.3)	84 (85.7)	0.23 (NS)	1.56	0.74–3.18	0.06 (NS)	2.29	0.94–5.45
grade V injury (massive splenic disruption): 257(26.9%)	51 (19.8)	206 (80.2)	<0.01	2.31	1.36–4.06	<0.001	3.10	1.61–6.19
Other spleen injury: 217 (22.7%)	21 (9.7)	196 (90.3)						
Associated injuries								
isolated BSI: 545(57.1%)	46 (8.4)	499 (91.6)	<0.05	0.59	0.39–0.90	–	–	–
compound IRR: mean (median)	0.75 (0.9)	0.85 (0.9)	<0.001	0.07	0.03–0.17	–	–	–
intracranial:37(3.9%)	10 (27.0)	27 (73.0)	<0.01	3.37	1.51–6.97	–	–	–
liver: 64(6.7%)	14 (21.9)	52 (78.1)	<0.01	2.48	1.28–4.55	–	–	–
hollow viscus: 23(2.4%)	11 (47.8)	12 (52.2)	<0.001	8.58	3.63–20.12	<0.001	11.03	3.46–34.28
pancreatic:20 (2.1%)	4 (20.0)	16 (80.0)	0.18 (NS)	2.16	0.61–6.03	–	–	–
other intraabdominal:127(13.3%)	23 (18.1)	104 (81.9)	<0.01	2.13	1.26–3.49	–	–	–
thoracic: 161(16.9%)	31 (19.2)	130 (80.8)	<0.001	2.47	1.54–3.89	–	–	–
upper limb: 99(10.4%)	17 (17.2)	82 (82.8)	<0.05	1.91	1.05–3.29	–	–	–
lower limb:42 (4.4%)	7 (16.7)	35 (83.3)	0.19 (NS)	1.74	0.69–3.81	–	–	–
spinal/back: 53(5.5%)	16 (30.2)	37 (69.8)	<0.001	4.16	2.17–7.67	0.07(NS)	2.14	0.90–4.92
pelvic fracture:46(4.8%)	10 (21.7)	36 (78.3)	<0.05	2.50	1.14–5.02	–	–	–
Management characteristics								
transfusion: N = 140(14.7%)	52(37.1)	88(62.9)	<0.001	9.24	5.91–14.51	<0.001	7.70	4.54–13.16
Era (reference new era post 2005)								
new era (January 2006–December 2011) N = 529(55.4%)	46(8.7)	483(92.3)						
old era (July 2000–December 2005) N = 426 (44.6%)	55(12.9)	371(87.1)	<0.05	1.56	1.03–2.36	0.13(NS)	1.47	0.89–2.43

The bold/italic values are the significant results.

lack of uniform paediatric trauma triage criteria and transfer guidelines dictate that many seriously injured children are cared for outside of PTCs, either at ATCs without specifically paediatric personnel or resources, or in combined centres of varying levels [47,48]. In one US multistate voluntary paediatric trauma database analysed recently, only one quarter (24.3%) of 9597 splenic injuries were cared for at a PTC [6]. The situation is similar in Canada, where 50–55% of injured children are cared for at non paediatric hospitals, and 38% in non-trauma centres [8,19]. In Victoria, another state with Australia, 83% of major paediatric trauma is reportedly managed in trauma centres but up to one third of this is outside of the state's single paediatric trauma centre [49].

The first NSW “State Trauma Plan” along with designated adult and paediatric trauma centres were introduced in the early 1990's, all predating this study. In 2002, the NSW Institute of Trauma and Injury Management (ITIM) was established. Prehospital retrieval services became increasingly coordinated over the ensuing decade and the most recent NSW Trauma Services Plan was released in

2009, designating seven major and ten regional/rural trauma centres with three PTCs [50].

Despite the existence of PTC's as well as primary retrieval and secondary transfer services for the duration of this study, we saw a similar situation to that in North America – almost two thirds of children with BSI were managed in non-paediatric centres, with almost 40% in ATCs, most of them rural, and 25% in non-trauma centres.

Admission to a paediatric centre (either primarily or after transfer) seems to depend on numerous factors, including proximity to the hospital, injury severity, any local triage guidelines and availability of and access to surgeons and retrieval services. A recent review of 59 international articles concerning the care of children in trauma systems found a lack of standard prehospital triage criteria [48]. Yet these seemingly ad hoc triage practices influence outcome. In both NSW and internationally, the hospital an injured child is admitted to has a significant impact on morbidity, mortality and functional outcomes. Children admitted

to a PTC have 3–5 times higher odds of having a survival advantage than if treated at an adult centre [7,48]. The operation rates for BSI in this study appear to be a microcosm of this finding.

It is reassuring that both in NSW and North America, the NOM rate is increasing. In NSW, rates rose both in and away from PTCs, although the rate in non PTCs of 88.3% remained just outside APSA guidelines [16]. Studies from North America have shown the same trends with NOM rates as low as 13% in the 1980's [38–40], climbing to as high as 94% by the end of the last decade [18,43,44]. Improvement, although not always meeting benchmarks, is mainly due to changes outside of PTCs [6]. Dissemination of clinical guidelines, publication of NOM rates, ongoing surveys, peer review and education programs [43] have all been credited with this success.

In 2010, ITIM recommended that primary transport of injured children to a PTC occur if feasible, otherwise initial assessment and stabilisation should occur at an ATC prior to secondary transfer [49]. It is hoped this recommendation will further improve the care of injured children in NSW. However, this recommendation needs to be supported with guidelines for any initial management in an ATC. These should include firm criteria for the triage of paediatric patients and their ultimate transfer to a PTC, along with the facilitation and promotion of NOM wherever a child with BSI is cared for.

The influence of patient factors

This study also confirmed the independent association of operation with a few specific patient factors including age, massive splenic disruption (but not other spleen injury grades) and blood transfusion. Other multicentre studies have concurred [6,18,21,29]. The association with transfusion is not surprising if one assumes this is a proxy for haemodynamic instability which is the one enduring indication for operation [26,37,51]. Of note, injury severity, as defined by C-IRR, and associated injury (aside from hollow viscus injury which often mandates a laparotomy) were not independently related to operation rate. This is not a universal finding, with higher injury severity [4,6,21,29], the presence of other or multiple injuries [6,18,21,23,26,29], and grade of splenic injury [18,23,24] previously correlated to the odds of operation. While we specifically didn't examine the effect of multiple injuries, the lack of effect of C-IRR (in which all injuries contribute to the severity measure) demonstrates that, aside from intestinal perforation, other injuries and injury severity certainly do not preclude NOM.

The influence of surgeon

An evaluation of the effect of surgical decision on BSI management was beyond the scope of this study, but others have identified this as a contributing factor. While most surgeons agree NOM is preferable in theory [33], in North America, operation rates have been found to be higher under adult trauma surgeons compared with paediatric surgeons, and conversely lower when management is by trauma surgeons with paediatric experience. Barriers surgeons identify include lack of paediatric resources and infrastructure, limited paediatric experience and educational updates, lack of appropriate surgical cover, lack of familiarity with clinical practice guidelines, use of adult algorithms, entrenched practices, concern about splenic injury grade, contrast blush on CT and other injuries, and low volumes of trauma [23,30–34,52,53]. In the absence of paediatric trauma triage criteria and clinical practice guidelines in NSW, it is likely that individual surgical decision making played a role in the OM rates identified in this study. This requires further evaluation.

Study limitations

Using administrative data is an efficient way to gain a broad picture of splenic injury and its management across the state but there is a potential for errors and omissions in data entry. We were unable to assign the more commonly used Injury Severity Score (ISS) to the ICD-10-AM so used C-IRR as an alternative measure. In addition, key factors in surgical decision making such as physiological parameters, other intraabdominal injuries and test results, cannot be directly assessed. Proxies including blood transfusion and C-IRR act only as a representation of these parameters. The number of deaths was small and any associations should be interpreted with caution. Insight into the system and individual decision making processes behind the data may enhance its interpretation. However, the consistency of our findings with other studies, including those that do include physiological data [26,50] and the correlation with what is known about surgical decision making, seems to validate the fact that there are unlikely to be other unknown differences between the populations managed at PTCs and non-PTCs in NSW.

Conclusion

This is the first study in Australia using administrative data to examine the management of BSI in children across the health system. The results suggest that children in NSW are being operated on for BSI unnecessarily, particularly away from paediatric centres. The factors at play are complex. Further evaluation is required, including examination of the management and movement of injured children within the broad NSW trauma system. In addition, review of the existence and use of practice and triage guidelines, as well as the hospital and surgeon practices in managing BSI, is warranted.

Conflict of interest

None.

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